

INFO 5601

Information Ethics and Policy

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	AI Engineer (63.0 / 100)	52.5 / 100

53	83	+38
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Explores ethical and legal complexities of information and communication technology. By combining real-world inquiry with creative speculation, students will probe everyday ethical dilemmas they face as digital consumers, creators and coders, as well as relevant policy. Explores themes such as privacy, intellectual property, social justice, free speech, artificial intelligence, social media and ethical lessons from science fiction.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	63.0	93.9	+30.9	HIGH
Machine Learning Engineer	61.1	89.3	+28.2	MEDIUM
ML Engineer	58.7	77.8	+19.0	MEDIUM
Applied Scientist	52.8	90.9	+38.2	HIGH
Data Scientist	47.9	85.1	+37.2	HIGH
MLOps Engineer	31.7	60.7	+28.9	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

Deep Learning

1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

AI Engineer

63.0 → 93.9 (+30.9 pts)

data robotics learning technical model systems teams based

Machine Learning Engineer

61.1 → 89.3 (+28.2 pts)

data web learning analytics tools solutions design development

ML Engineer

58.7 → 77.8 (+19.0 pts)

data learning machine systems model world problems impact

Applied Scientist

52.8 → 90.9 (+38.2 pts)

research interdisciplinary problems technical data learning systems industry

Data Scientist

47.9 → 85.1 (+37.2 pts)

definition data analytics people problems approaches

MLOps Engineer

31.7 → 60.7 (+28.9 pts)

data model systems learning machine training tools development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.